

PAQ: Prefill-Once, Ask-Many

Reversible, query-conditioned page selection for multi-query document VQA

A living research document — current state, honest negatives, and roadmap
(status: 2026-07-16; this document tracks ongoing work and is updated as results land)

BASED-MM program

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Abstract

Real document-VQA benchmarks are *multi-query*: one long document is prefilled once and asked many different questions (in our pinned evaluation sets, 5.5–5.8 gold-bearing questions per document). Standard KV-cache eviction commits to *one* compressed context and reuses it for every question. We study PAQ (“Prefill-once, Ask-many”): retain the coarse document prefill and, *per query*, re-select a small page subset with the reader’s own question→page attention, then reread those pages at full resolution. The axis we test is **reversibility** — per-query re-selection versus commit-once selection — not query-awareness.

What is confirmed. On a 903-query confirmatory run (80 documents per cell, both cells, document-clustered inference), per-query re-selection beats *the identical scorer committed once* (pooled accuracy +0.210, 95% CI [+0.168, +0.250]) and a strong gold-informed static subset (+0.106, [+0.067, +0.141]) — on both cells, surviving an independent-extension check on the 50 new documents per cell and a worst-case treatment of 4 timeout-missing queries. The same fresh-over-stale pattern also holds for the retrieval scorer tested here (ColQwen2 fresh 0.423 vs. stale 0.178 pooled). A KV-cache reuse bug later found in the per-query scoring loop (§4.3) was quantified and the isolation contrast re-verified with the fix in place (clean +0.264, LB₉₅ = +0.187, vs. polluted +0.284 on a 24-document check): the reversibility claim stands.

What is equally true and reported with the same prominence. As a page selector, PAQ *loses* head-to-head to a strong trained visual retriever: ColQwen2 re-ranked per query beats PAQ by ≈ 10 accuracy points at matched page budget (pooled -0.099 , 95% CI $[-0.129, -0.072]$), on every reported stratum. Moreover, CAPS (ACL 2026) has already published attention-based page selection beating ColPali-family retrieval on these same benchmarks — the page-level *method* is scooped. An expert-head variant (PAQ-T) reaches *recall* parity with ColQwen2 (clean gate: 0.683/0.626 vs. 0.664/0.536 per cell), and its accuracy read-eval has now run: PAQ-T is **non-inferior** to ColQwen2-fresh at matched budget (near-full coverage: pooled +0.017, LB₉₅ = -0.000 ; per cell +0.006/+0.028) — *parity* with a SOTA trained retriever, training-free, with no separate model, but **not a beat**. Its head is selected by cross-validation on these benchmarks’ gold (benchmark-supervised, not zero-shot), and expert-head selection is CAPS’s technique.

What this document is. An honest consolidation of an ongoing project: the confirmed multi-query reversibility result, the first *executed* retrieval-vs-selection panel on MMLongBench-Doc/LongDocURL at 64K context, the PAQ-T accuracy-parity result, and a two-lane program now under way to convert that parity into a beat: (B) an off-benchmark LoRA selection-adaptor (designed, cross-family reviewed, CPU preparation done, GPU-gated) and (C) below-page PAQ-KV reversible KV-block selection (designed, decode-identity spike passed, 30-document development screen in progress at the time of writing). Both lanes are honestly ~ 15 – 25% bets; a sealed fresh test set (73/45 documents, untouched) is reserved for a single final evaluation. They are work in progress, not results.

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1 Introduction

1.1 The problem

Long-context vision-language models (VLMs) can read an entire multi-page document, but they pay for every page on every question. The standard efficiency response, inherited from single-query text benchmarks, is to *evict*: keep a compact subset of the KV cache (SnapKV, H2O, FastV, LOOK-M, KVzip) and decode against it (Li et al., 2024; Zhang et al., 2023; Chen et al., 2024; Wan et al., 2024; Kim et al., 2025). Eviction is a *single, irreversible commitment*: whatever subset is kept must serve every future question about that document.

1.2 The multi-query regime

Document assistants are not asked one question. The two long-document VQA benchmarks we use are natively multi-query: grouping the 64K splits by source document, MMLongBench-Doc (Ma et al., 2024) averages ≈ 6.97 questions per document and LongDocURL (Deng et al., 2025) ≈ 3.6 ; inside our pinned evaluation subsets (which require ≥ 3 gold-bearing questions per document) the averages are 5.50 and 5.79 respectively (Table 1). Different questions need different pages, so a subset committed for question 1 is *stale* for question 5.

1.3 The question this project asks

*At a matched page budget on long multimodal documents, does per-query **re-selection** of context beat a **commit-once** selection — and can a training-free, reader-internal selector compete with a dedicated trained retriever?*

The answer, on the evidence in this document, is split, and we state it up front:

- **Yes** to reversibility: re-selection beats commit-once, robustly, on both cells, for our scorer *and* for a retrieval scorer (§7.1, §7.2). This is the confirmed contribution.
- **No, at page level, to a beat**: PAQ v1 loses to ColQwen2-fresh (Faysse et al., 2025) by ≈ 10 accuracy points (§7.2), and CAPS (Zhu et al., 2026) already published the page-level attention-selection method (§8.1). Cross-validated expert-head scoring (PAQ-T) closes the *recall* gap and its accuracy read-eval now lands at **non-inferiority (parity)** with ColQwen2-fresh (§7.4) — a real property (training-free, single model), but not the beat this program is mandated to deliver.
- **Open** below page granularity and via calibration: two lanes are running to convert parity into a beat — an off-benchmark LoRA selection-adaptor (§4.6) and reversible below-page KV-block selection (§4.5; decode-identity spike passed, development screen in progress). Whether either improves end-to-end accuracy is untested; the roadmap remains explicitly speculative (§9).

1.4 Reading guide

§2 fixes benchmarks and pins; §3 the executed baseline panel; §4 the methods — including a methods-integrity note on a KV-cache bug found and fixed mid-program (§4.3), the executed PAQ-T accuracy read-eval protocol (§4.4), and the two new lanes under test (§4.5, §4.6); §5 the statistical machinery; §6 the algorithms; §7 all results (positive and negative, with per-cell results primary and pooled results explicitly labelled); §8 related work including the CAPS overlap; §9 current

Table 1: The confirmatory evaluation pin (the numbers that should be cited). “Complete queries” are rows where every gated arm scored (complete-case pairing); 4 of 907 vision queries are missing due to a per-document timeout (§7.1).

	LongDocURL	MMLongBench-Doc
Documents	80	80
Complete queries at gate ($b=5\%$)	463	440
Gold-bearing queries / document (pin)	5.79	5.50
Median (mean) union pages / document	44 (45.0)	28 (36.5)
Benchmark-level questions / document (64K split)	≈ 3.6	≈ 6.97

state and roadmap; §10 limitations. The appendix carries the full panel, the adversarial review trail, and reproducibility pins.

2 Benchmarks and setup

2.1 Cells, documents, pins

We evaluate on two cells: **LongDocURL** (Deng et al., 2025) and **MMLongBench-Doc** (Ma et al., 2024), both consumed as page images (no OCR). For each document we build the *union of the benchmark’s own-document pages* across that document’s queries (dropping cross-document 64K padding), so every query’s evidence is present in one prefill. Documents qualify if they have ≥ 3 gold-bearing queries and ≤ 110 union pages ($\approx 64K$ tokens at read resolution); selection is seed-0 frozen.

Two pins exist: the 30-document/cell *screen* pin and the 80-document/cell *confirmatory* pin. The screen pin is a strict prefix of the confirmatory pin (same seed, same arrival order), which is why §7.1 additionally reports an *independent-extension* analysis on the 50 *new* documents per cell — the confirmatory is not independent of the screen unless restricted to those. Both pins are sha-256-checksummed and the harness verifies the screen pin byte-identically before use (Appendix C).

2.2 Reader, budgets, metric

The reader is **Qwen3-VL-8B** (Qwen Team, 2025), frozen, eager attention. All selection arms operate at a matched *page* budget $b \in \{2.5\%, 5\%, 10\%\}$ of the document’s pages (the pre-registered gate budget is $b = 5\%$); the selected pages are re-rendered at full resolution and read. Accuracy is the benchmark’s document-VQA scorer (exact/relaxed match by answer format). Scope note: everything in this document is a **Qwen3-VL-8B case study**; no second VLM has been run (§10).

2.3 The base-model ceiling

A large fraction of queries receive no credit from this reader even when it is given *exactly* the annotated gold-evidence pages: on the confirmatory run the oracle-evidence arm scores 0 on $317/903 = 35.1\%$ of queries and scores a full 1 on only $407/903 = 45.1\%$ (the pre-specified “answerable subset”). This limits absolute scores. Pairing controls for shared query difficulty, but does not imply that the ceiling affects every selector equally; we report the answerable subset only as a secondary view because it conditions on a model outcome. The oracle row is *not* a strict ceiling: PAQ beats oracle-evidence on 28 queries, because a selected page set can differ from the annotated gold set in ways that help the reader.

Table 2: Arm taxonomy. The load-bearing axis is **reversible** (re-select per query) vs. **commit-once** (one subset per document, reused for all its queries); the second axis is deployability. Disclosed proxies are marked.

Arm	Reversible?	Deployable?	Role
PAQ (ours)	yes	yes	the method
attn_max (ours, max-agg)	yes	yes	reported alternate aggregation
paq_stale	no	yes	isolation: same scorer, committed once
gold_frequency_static	no	no (uses gold)	isolation: strong gold-informed static
ColQwen2-fresh	yes	yes	strong deployable retrieval , re-ranked per query
ColQwen2-stale	no	yes	retrieval committed once (same commit policy)
CLIP-fresh	yes	yes	cheaper retrieval control (Radford et al., 2021)
SnapKV-fresh	yes	yes	reference (weak port; beating it not required)
SnapKV-stale	no	yes	strongest stale family arm
H2O / FastV / LOOK-M / KVzip	no	yes	demoted query-blind controls (proxies, see text)
random / truncation	no	—	controls
full-context / oracle-evidence	—	—	reference rows, <i>not</i> competitors

3 Baselines: the executed panel

Every arm below was actually run, per query, at matched page budget, on the same pinned data with the same reader. The resulting panel directly compares retrieval and reader-internal selection on MMLongBench-Doc/LongDocURL at 64K in the multi-query regime; our current literature search did not identify an earlier executed panel with this exact scope. Table 2 gives the taxonomy; Appendix A the full numbers.

The demoted eviction family (disclosed proxies). H2O uses accumulated received attention; FastV is a *context-only proxy* (canonical FastV is prompt-conditioned); LOOK-M degenerates to H2O on image-only pages; KVzip is a *recv_max* proxy for the reconstruction pass of Kim et al. (2025). On these cells the whole family sits *at or below random* page recall, so beating it is not a claim we make — they are query-blind controls, retained for completeness, and a faithful-reconstruction KVzip would likely be stronger. The honest comparators are *paq_stale*, *gold_frequency_static*, SnapKV-stale/fresh, and the retrieval arms.

On gold_frequency_static. This arm keeps the top- B pages by *gold-page frequency* across the document’s queries. It uses gold labels, so it is not deployable; it is a *strong gold-informed static commitment* — but it is **not** provably accuracy-optimal and we do not call it “best possible” or a “ceiling” (an earlier draft did; the claim was retracted under review, Appendix B).

Reference rows. Full-context and oracle-evidence are reported for calibration only. SnapKV-fresh (re-run per query) is a reference, not an upper bound; our port averages heads and layers before pooling and is weaker than canonical SnapKV.

4 Methods

4.1 PAQ: coarse-to-fine page routing

PAQ is training-free page routing over a frozen VLM:

1. **Prefill once (per document).** Pages are rendered at coarse resolution ($\approx 3.5\text{K}$ visual tokens total, i.e. ≈ 70 tokens/page on a 50-page document) and prefilled once; the KV cache is retained. Attention statistics are captured by OOM-safe per-layer forward hooks (Algorithm 4).
2. **Re-select per query.** The question is appended to the retained cache; the question rows’ attention over page tokens (mean over heads, layers, and question tokens) is averaged over each page’s token span to give a per-page score (Algorithm 1). Keep the top pages at budget b .
3. **Reread.** The selected pages are re-rendered at full resolution and the question is answered from them.

Scope, stated plainly: this is *page routing* — coarse-score $\rightarrow$ pick pages $\rightarrow$ full-resolution reread. It is *not* token-level KV compression: we never decode from a masked retained cache, and we claim no memory savings in v1. The estimand is query-conditioned page selection at matched page budget.

4.2 The reversibility framing and the isolation protocol

“Query-aware selection beats query-agnostic eviction” is both a known text result (Tang et al., 2024) and a strawman risk (FastV and LOOK-M can attend to the question). We therefore fix the axis to **reversibility**: every baseline runs its canonical scorer but is deployed *stale* — committed once per document (query-aware members commit an aggregate over the document’s first $M = \min(3, k)$ queries) — while PAQ re-selects per query (Algorithm 2).

Reversibility is credited *only* if PAQ beats **both**

- **paq_stale:** PAQ’s *own* scorer committed once. Identical scorer, identical budget; the only difference is reversibility, so the delta isolates the mechanism;
- **gold_frequency_static:** the strong gold-informed static, so the win is not an artifact of a weak committed selector.

This gate exists because a first, weaker gate (PAQ vs. best-of-eviction-family) *passed and was then retracted*: adversarial review showed the family’s stale arms were weak-static artifacts (a gold-informed static nearly matched PAQ in recall), so “reversibility” had not been isolated. The two-comparator gate above was frozen before the confirmatory run (Appendix B documents both review rounds).

4.3 Methods integrity: the KV-cache pollution bug and its fix

A cross-family build review of the PAQ-T read-eval harness (2026-07-16) found a real bug in the per-query scoring loop used across this program: **transformers 5.x mutates the retained *DynamicCache* in place**, and the loop reused the document prefill across a document’s queries *without* restoring it (**past.crop(n_ctx)**) after each question forward. Query q_i therefore scored while attending to $q_0 \dots q_{i-1}$ in addition to the document. The bug was present in the PAQ-T recall gate *and* in the confirmatory scorer (**paq.py**, **HookedBackend.query_scores**) — i.e. in the run behind §7.1. The fix is the crop call after every per-query preview; it is now applied everywhere.

Its impact was quantified rather than assumed:

- **Blast-radius diagnostic** (`paq_kvcrop_diag.json`; 6 docs/cell, 87 query-rows): the bug is not inert — for a *fixed* scoring head it inflates gold-page recall by $\approx +0.05$ and shifts selections (mean top- B Jaccard clean vs. polluted 0.82), with the drift growing in the query index (Jaccard $1.00 \rightarrow 0.61$ from q_0 to q_6), as expected for cumulative cache contamination.
- **The recall gate was regenerated CLEAN** before any accuracy read: still PAQ_T_VIABLE; held-out expert recall 0.683/0.626 (vs. polluted 0.674/0.628), and the clean CV picks a more robust head (layer 21). All PAQ-T numbers in this document are from the clean gate.
- **The reversibility result was re-verified against the fix (§7.1)**: on a 24-document/150-query clean-vs-polluted re-run, PAQ-paq_stale is +0.264 ($LB_{95} = +0.187$) clean vs. +0.284 polluted, and PAQ-gold_frequency_static +0.187 ($LB_{95} = +0.137$) clean vs. +0.228 polluted. The isolation contrast largely cancels the bug because both arms share the same (equally polluted) scorer; absolute PAQ accuracy in the confirmatory does carry some inflation (clean-polluted -0.041 on this check), which we disclose wherever absolute levels matter. The confirmatory reversibility *claim* stands.

Lesson recorded for the program: this is the second in-place-cache bug this codebase family has hit; every per-query preview over a retained cache now crops, and GPU smoke tests exercise the full path.

4.4 PAQ-T: expert-head scoring and the executed accuracy read-eval

After the retrieval loss (§7.2) we tested how much of the gap is an artifact of PAQ’s mean-aggregation and resolution starvation (≈ 70 coarse tokens/page vs. ColQwen2’s $\approx 1,024$ patch embeddings/page). PAQ-T scores with (a) more scoring tokens per page (256 tokens/page, 6,000-token target total — a cap forced by encoder OOM on large documents, not full page resolution), and (b) a single *expert (layer, head)* chosen by held-out-document cross-validation (Algorithm 3) rather than the mean over all heads. Expert-head selection is CAPS’s technique (Zhu et al., 2026); what is ours here is the amortized multi-query architecture it plugs into. “Training-free” means no parameter updates.

Claim scope (stated before the numbers). The head is chosen using gold page-recall on the complementary document fold *of the same benchmarks*. Cross-fitting removes row-level leakage (no query is scored by a head selected on its own document’s fold), but this is **cross-validated, benchmark-supervised model selection** — **explicitly not zero-shot**. Removing this caveat is precisely the job of the B-lane adapter (§4.6).

The accuracy read-eval protocol (executed 2026-07-15/16; bars frozen before the run). The clean recall gate (§4.3) selects the expert heads: fold heads 21_11/21_18 on LongDocURL and 21_11/21_11 on MMLongBench-Doc — all in layer 21. Each held-out query is scored with the head trained on the *disjoint* fold (no selection-on-test); its top- B pages at $b = 5\%$ are then re-rendered at full resolution and read by the **same** sdpa reader, render policy, and scorer as the cached ColQwen2-fresh baseline rows — only the selected page set differs between arms. Inference is the document-cluster paired bootstrap of §5. Pre-registered bars (`paq_t_readeval_frozen_meta.json`): NONINFERIOR iff both cells’ $LB_{95}(\text{PAQ-T-ColQwen2-fresh}) > -0.03$; BEATS iff pooled and both per-cell $LB_{95} > 0$; INFERIOR iff either cell $LB_{95} \leq -0.03$; per-cell power floors (150 queries, 15 documents) and an adversarial missingness imputation are part of the verdict. Because the high-resolution scoring pass OOMs on a handful of very large documents, coverage was completed by a *conservative* recovery

pass: the missing documents were re-scored at a reduced 4,000-token scoring target — degrading only the PAQ-T arm’s selection, never the cached baseline — bringing coverage to 463/463 and 440/444 (§7.4). Result: non-inferiority (parity), not a beat.

4.5 PAQ-KV: reversible below-page KV-block selection (C lane, being tested)

PAQ-KV pushes the confirmed reversibility mechanism *below* page granularity — the one operating point our literature search still finds unoccupied (§8). The document is prefilled **once** at the same rendering the cached full-context arm used ($\leq 64K$ tokens) and the KV cache is retained. Then, per query:

1. **Score:** the reader’s own question→token attention (the same crop-fixed preview used everywhere in this program) scores 64-token, page-aligned KV blocks;
2. **Select:** keep the top blocks at a **nominal-5% decode-attention budget** (terminology deliberate: this is *not* a matched-token comparison — selected blocks’ K/V encode full-document prefill context, the standard claim semantics of the Quest/SnapKV method class, and the realized fraction in the spike is $\approx 5.7\text{--}6.1\%$ of context);
3. **Decode:** generate the answer attending *only* to the selected blocks plus a small always-keep set (BOS/system/template scaffold, the question, generated tokens), via a custom **4D additive attention mask over the retained cache — no gathering** — so M-RoPE positions are exactly the model’s own (`get_rope_index` unchanged). No re-read, no re-render, no second model. Selection is fully reversible: the next query re-selects from the same complete cache.

The mask route is the *accuracy* path; physically gathering the cache (for bandwidth/memory savings) is a separate, future efficiency claim that we do not make. A decode-identity audit validated the mask machinery exactly: over 6 document–query pairs (32K–122K tokens), an all-true mask reproduces the native causal decode with **zero mask-specific argmax changes** (`paqc_identity_audit.json`); the earlier spike’s single greedy-token divergence traced to **generate’s** mask-independent **is_causal** fast-path kernel, not to the mask. PAQ-KV is distinct from CAPS (a *separate* scoring VLM, per-page full re-forwards, single-query) and from page-PAQ (page-quantized re-reading): it never re-reads, and its selection unit is below the page. Status: a 16-query two-cell spike showed PAQ-KV@5% $\approx$ full-context on both cells; a 30-document/cell development screen (with commit-once, wrong-query, random-block, page-attend, and 2-D tile controls, and frozen STOP gates) is **running at the time of writing** — no screen results are reported here.

4.6 PAQ-T-LoRA: an off-benchmark selection-adaptor (B lane, planned)

The B lane attacks the parity result’s two weaknesses at once — no beat, and benchmark-supervised head selection. The method is a **toggleable LoRA on the reader’s expert-layer Q/K projections**: enabled *only* for the scoring forward (the question preview over the coarse document prefill), **disabled for the answer-generation read**, so a single deployed model both selects and reads and the reading model is byte-identical to the frozen baseline (an invariance test is part of the plan). The adapter is trained contrastively on **off-benchmark** data — MP-DocVQA validation (page-labeled, image-native), with every training page provenance-checked (perceptual hash + source identity) against all 40,442 benchmark page images — and *everything* (expert layer/heads, rank, temperature, checkpoint, seed, resolution policy) is selected on external development data; the benchmark-picked heads of §4.4 are demoted to diagnostics. The goal is to convert selection

parity into a beat by shifting queries out of ColQwen2’s selection-win bucket (§9.3). Status: designed and twice cross-family reviewed (verdicts REVISE/HOLD, then HOLD pending frozen splits — all fixes folded); CPU preparation is done (sealed fresh evaluation set, provenance index, per-head mixture harness); all GPU stages are gated. No training has run; there are no B-lane results.

5 Definitions and statistical machinery

Matched page budget. For a document with n pages and budget fraction b , every arm keeps

$$B = \max(1, \lfloor b \cdot n \rfloor) \quad (1)$$

pages. Pages have unit budget weight in the executed vision harness, so this is the exact behavior of its cumulative-budget selector; ties in score are broken by page index. The same selector code path is used for every arm.

Gold-page recall at budget. For query q with nonempty annotated gold-page set G_q and kept set K_q ($|K_q| = B$),

$$\text{recall@}b(q) = \frac{|K_q \cap G_q|}{|G_q|}. \quad (2)$$

Reported recall is the query-macro mean of this fraction within a cell; it is not answer accuracy and it is not the binary event “at least one gold page retrieved.”

Commitment pressure. For a document with g distinct gold pages across its k queries,

$$\text{pressure} = \frac{g}{b \cdot n}, \quad (3)$$

i.e. how many times over the b -budget capacity the document’s queries collectively need. This pre-registered quantity uses the fractional, pre-rounding capacity bn , whereas the executed selector uses B from Eq. (1). Thus pressure > 1 means $g > bn$; when $bn \geq 1$ this also implies $g > B$ and no single B -page subset can contain every distinct annotated gold page. For $bn < 1$ and at rounding boundaries, pressure should be read as a continuous diagnostic rather than a literal coverability test. This was the pre-registered mechanism axis; the formal test of it **failed** (§7.3) and it is retained as exploratory only.

Document-cluster paired bootstrap and LB₉₅. For arms A and C , let $d_q = s_A(q) - s_C(q)$ be their paired accuracy-score difference on query q . With two cells $c \in \{1, 2\}$ and N_c queries in cell c , the pooled point estimate is

$$\hat{\Delta}_{A-C} = \frac{1}{2} \sum_{c=1}^2 \left(\frac{1}{N_c} \sum_{q \in c} d_q \right). \quad (4)$$

Queries of one document share the document *and* the single stale subset, so bootstrapping over queries would be anti-conservative pseudoreplication. All intervals therefore resample *documents* with replacement within each cell (10,000 draws, seed 0), recompute Eq. (4), and take percentile endpoints. Per-cell intervals use the same procedure with one cell. We define

$$\text{LB}_{95}(\Delta) = 2.5\text{th percentile of the bootstrap distribution of } \Delta, \quad \text{UB}_{95}(\Delta) = 97.5\text{th percentile}, \quad (5)$$

i.e. LB₉₅ is the **lower end of a two-sided 95% CI** (equivalently a one-sided 97.5% bound). An earlier label of “one-sided 95%” was incorrect and has been fixed; the frozen gate arithmetic never changed.

Algorithm 1 PAQ per-query scorer and matched-budget selection (vision cells)

Require: document pages $P_1, \dots, P_n$; queries $q_1, \dots, q_k$; budget b ; frozen VLM (eager attention)

- 1: render all pages coarsely ($\approx 3.5K$ visual tokens total); prefill once; retain KV cache C
- 2: record image-token spans $s_1, \dots, s_n$ for the pages
- 3: **for** each query q_j **do**
- 4: forward only the tokenized question against C (no re-prefill)
- 5: via hooks: $\mathbf{a} \leftarrow$ mean over layers, heads, and question rows of attention to each context position ▷ Algorithm 4
- 6: per-page score $\text{score}_j(p) \leftarrow |s_p|^{-1} \sum_{t \in s_p} \mathbf{a}[t]$ ▷ **attn_max** variant: max over s_p
- 7: $K_j \leftarrow$ top- B pages by score_j , $B = \max(1, \lfloor bn \rfloor)$ ▷ Eq. (1); index tie-break
- 8: re-render pages K_j at full resolution; answer q_j from them
- 9: **end for**

Algorithm 2 Commit-once (stale) vs. re-select (fresh) deployments — the reversibility contrast

Require: per-query score vectors $v_1..v_k \in \mathbb{R}^n$ from ANY scorer (PAQ, SnapKV, ColQwen2)

- 1: **fresh (reversible):** for query q_j , select top- B from v_j ▷ one subset per query
- 2: **stale (commit-once):** $u \leftarrow \frac{1}{M} \sum_{j \leq M} \text{minmax}(v_j)$ with $M = \min(3, k)$; select top- B from u once; reuse that subset for ALL k queries
- 3: *min-max normalization before averaging prevents any single query’s magnitude from dominating the committed ranking; query order is frozen (arrival order).*
- 4: *query-agnostic arms (H2O/FastV/LOOK-M/KVzip/random/trunc) are stale by construction (their score vector never sees a question).*

The isolation gate (pre-registered, frozen before the run).

$$\text{ISOLATED} \iff \text{LB}_{95}(\hat{\Delta}_{\text{PAQ-paq_stale}}) > 0 \ \wedge \ \text{LB}_{95}(\hat{\Delta}_{\text{PAQ-gold_frequency_static}}) > 0, \quad (6)$$

using complete cases at $b = 5\%$. The frozen gate is pooled with equal cell weighting; the per-cell intervals are corroborating analyses, not additional gate clauses. If either interval includes zero, the evidence is insufficient to isolate a reversibility benefit; that outcome would not establish equivalence between commit-once and per-query selection.

6 Algorithms

7 Results

The primary confirmatory numbers below come from `research/paq_verdict.json` (903 complete vision queries, 80 documents/cell, frozen v3 harness). The independent-extension, mechanism, and fusion diagnostics come from the documented post-hoc scripts in Appendix C; PAQ-T recall numbers come from the *clean*, *crop-fixed* `research/paq_t_gate_verdict.json` and PAQ-T accuracy numbers from `research/paq_t_readeval_verdict.json` and `research/paq_t_readeval_combined_verdict.json`. Per-cell results are primary. “Pooled” inferential contrasts use the equal-cell statistic in Eq. (4); query-micro summaries are labelled separately.

Algorithm 3 Expert-head selection with held-out-document CV (PAQ-T; gold-supervised selection on the complementary fold, never on the scored documents)

Require: per-(layer,head) question→page scores; gold sets G_q ; documents D of one cell; budget $b = 5\%$

- 1: split D into two folds by document (seed-0 shuffle)
 - 2: **for** each fold F **do**
 - 3: $Q_{\text{train}} \leftarrow$ queries in $D \setminus F$
 - 4: on Q_{train} : pick $(\ell^*, h^*) = \arg \max_{\ell, h} |Q_{\text{train}}|^{-1} \sum_{q \in Q_{\text{train}}} \text{recall}@b_{\ell, h}(q)$ ▷ Eq. (2)
 - 5: evaluate (ℓ^*, h^*) on the held-out fold F
 - 6: **end for**
 - 7: report query-macro held-out recall per cell ▷ no weight updates; head choice does use training-fold gold labels
-

Algorithm 4 OOM-safe per-layer-hook attention capture (the engineering that makes 64K scoring feasible)

- 1: register a forward hook on every language-decoder self-attention module (vision encoder excluded)
 - 2: run the chunked prefill / question forward with `output_attentions=False` under eager attention
 - 3: inside each hook: catch the layer’s $(1, H, q, k)$ attention tensor, reduce it immediately (fp32-accumulated column-sum for H2O, running max for KVzip, layer-2 last row for FastV, question-row mean for PAQ), and **discard** it
 - 4: peak memory = ONE layer’s attention. The naive all-layers `output_attentions=True` path OOMs (≈ 77 GB at 128K text) or crashes vision cuDNN.
 - 5: a capture guard hard-fails if no hook observed a 4D tensor (prevents silently scoring zeros)
-

7.1 Confirmatory isolation: reversibility is real

CONFIRMED Per-query re-selection beats the identical scorer committed once AND a strong gold-informed static, on both cells, document-clustered, surviving independent-extension and missingness worst-case checks.

Table 3 and Figure 1 give the gate result. At $b = 5\%$:

At screen scale (30 documents/cell), the MMLongBench-Doc contrast against the static was -0.017 with a CI including zero. It became positive at 80 documents; we treat that earlier result as an underpowered screen, not as an independent replication.

Independent extension (the screen is embedded in the confirmatory). Because the 80-document pin contains the 30 screen documents, we re-ran the isolation statistics on the **50 new documents per cell only** ($n = 582$ queries): PAQ – `paq_stale` $+0.216$ [$+0.159, +0.269$]; PAQ – `gold_frequency_static` $+0.108$ [$+0.059, +0.150$]; per-cell static contrasts $+0.089$ [$+0.004, +0.162$] (LongDocURL) and $+0.127$ [$+0.083, +0.173$] (MMLongBench-Doc). The result therefore also holds on the out-of-screen extension.

Answerable subset (secondary). On queries where oracle-evidence scores exactly 1 ($n = 407$, 137 docs): PAQ 0.583 vs. `paq_stale` 0.182 ($+0.395$ [$+0.324, +0.460$]) and vs. the static 0.382 ($+0.204$ [$+0.135, +0.266$]). Reported secondarily because it conditions on a model outcome.

Table 3: Confirmatory isolation at $b=5\%$ on 903 complete queries (accuracy; document-cluster paired bootstrap). Contrast rows report $\Delta [LB_{95}, UB_{95}]$; the pooled column uses equal cell weighting.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
PAQ accuracy	0.385	0.258	0.322
paq_stale accuracy	0.128	0.095	0.111
gold_frequency_static accuracy	0.278	0.154	0.216
PAQ – paq_stale	+0.257 [+0.184, +0.330]	+0.163 [+0.126, +0.200]	+0.210 [+0.168, +0.250]
PAQ – gold_freq_static	+0.108 [+0.045, +0.164]	+0.104 [+0.063, +0.144]	+0.106 [+0.067, +0.141]
Verdict	REVERSIBILITY ISOLATED (both $LB_{95} > 0$, both cells and pooled)		

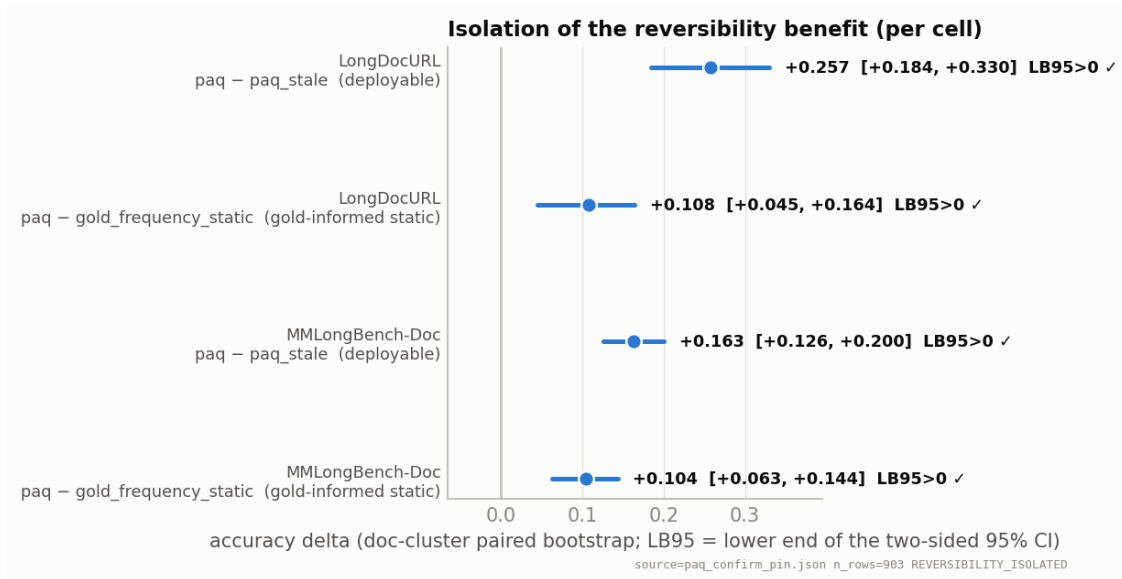


Figure 1: The statistical headline: per-cell forest of the two isolation contrasts at $b = 5\%$. Intervals are document-clustered two-sided 95% CIs; all four exclude zero.

Missingness. 4 of 907 vision queries (MMLongBench-Doc document `mi_phone`, queries 2–5) are missing due to a per-document timeout. Imputing them worst-case in either direction leaves the gate verdict unchanged.

Robustness to the KV-cache fix (integrity check, 2026-07-16). The confirmatory run predates the discovery of the cache-pollution bug (§4.3), so the isolation contrast was re-verified with the fix in place: on 24 documents/150 queries re-run clean vs. polluted at the confirmatory’s coarse configuration (`research/paq_revcrop_verdict.json`), PAQ–paq_stale is **+0.264** [$LB_{95} = +0.187$, $UB_{95} = +0.334$] *clean* vs. +0.284 *polluted*, and PAQ–gold_frequency_static is +0.187 [+0.137, +0.246] *clean* vs. +0.228 *polluted*. Because PAQ and paq_stale share the same scorer, the pollution largely cancels in the contrast; the bug inflated the headline delta by only ≈ 0.02 . Absolute PAQ accuracy is inflated by ≈ 0.04 on this check (*clean*–*polluted* -0.041 , $[-0.077, -0.003]$), which should be kept in mind when reading the absolute (non-contrast) rows of Table 3. **The confirmatory reversibility claim stands under the fix.**

Accuracy at matched page budget (per cell — not averaged)

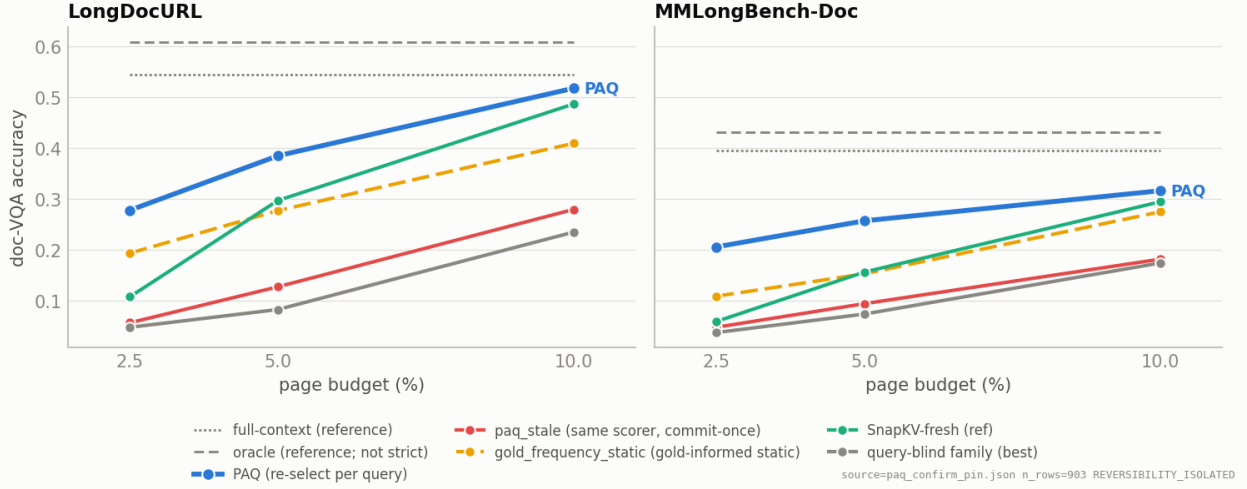


Figure 2: Accuracy at matched page budget, per cell (never averaged). Full-context and oracle-evidence are recessive reference lines, not competitors; the “query-blind family (best)” line is the demoted control family’s best member.

Budget curves. Figure 2 shows accuracy at $b \in \{2.5, 5, 10\}\%$: PAQ dominates **paq_stale** and the static at every budget on both cells, monotonically. On LongDocURL, PAQ at $b = 10\%$ reaches 0.518 vs. full-context 0.546 — reading 10% of pages recovers $\approx 95\%$ of full-context accuracy on that cell (not on MMLongBench-Doc: 0.317 vs. 0.397).

7.2 The deployable retrieval comparison: PAQ loses at page level

CONFIRMED NEGATIVE A strong trained visual retriever re-ranked per query beats PAQ by ≈ 10 accuracy points at matched page budget. This is the decisive result of the final review round, reported with the same prominence as the positive.

The adversarial AAI-readiness review named one decisive missing experiment: a deployable query-conditioned *retrieval* baseline. We ran it: ColQwen2 (Faysse et al., 2025) (late-interaction MaxSim over page images; pages embedded once per document) fresh and stale, plus CLIP (Radford et al., 2021) fresh, same reader, same budgets, all 903 queries. At $b = 5\%$, pooled, document-clustered (Table 4, Figure 3):

Three readings, all of which we stand behind:

1. **PAQ is not a superior page router.** The loss is ≈ 10 points, LB_{95} well below zero, and it holds on every stratum we examined (single-gold -0.110 , multi-gold -0.086 ; no question type where attention beats retrieval).
2. **The reversibility pattern spans both tested scorer families.** On the query-micro summary, ColQwen2-fresh is 0.423 vs. 0.178 stale; the corresponding equal-cell means are 0.420 and 0.177. This reproduces the fresh-vs-stale collapse with a different scorer. It supports per-query reselection for attention- and retrieval-based systems in this evaluated regime; it is not a claim about every possible scorer.

Table 4: Deployable retrieval comparison at $b=5\%$ (accuracy; 903 complete queries). PAQ loses to ColQwen2-fresh on both cells. Fresh exceeds stale for both tested scorers. The pooled arm means use equal cell weighting; contrast intervals are document-clustered.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
ColQwen2-fresh	0.512	0.328	0.420
PAQ	0.385	0.258	0.322
ColQwen2-stale (commit-once)	0.216	0.138	0.177
CLIP-fresh	0.155	0.164	0.159
full-context (reference)	0.546	0.397	—
PAQ – ColQwen2-fresh	pooled -0.099 $[-0.129, -0.072] \Rightarrow$ PAQ loses		
PAQ – ColQwen2-stale	pooled $+0.145$ $[+0.101, +0.185]$		
PAQ – CLIP-fresh	pooled $+0.162$ $[+0.128, +0.197]$		

Deployable retrieval comparison @ $b=5\%$ — PAQ loses to ColQwen2-fresh; reversibility (fresh > stale) holds for BOTH scorers

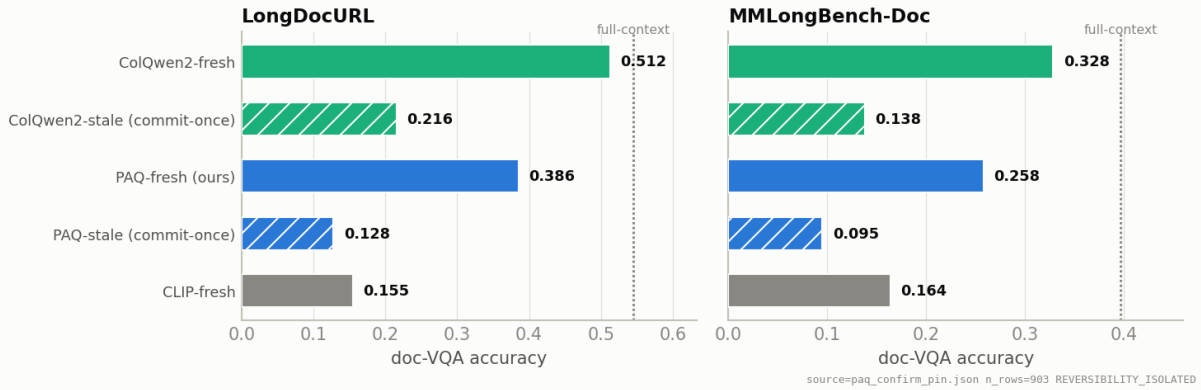


Figure 3: The two honest facts in one picture, per cell: (i) ColQwen2-fresh beats PAQ (the ≈ 10 pt loss); (ii) *both* scorers collapse when committed once — reversibility is a general property of the multi-query regime, not a PAQ-specific effect.

3. **The two existing outputs have limited complementarity.** A per-query oracle taking the better observed answer score from {PAQ, ColQwen2-fresh} gains $+0.035$ pooled over ColQwen2-fresh ($LB_{95} +0.024$; $+0.041/+0.030$ per cell). Among the 419 queries where ColQwen2-fresh scores exactly zero, PAQ scores one on only $\approx 6\%$. This makes simple output routing low priority. It does *not* upper-bound a fusion or reranker that selects a new page set, whose reader output could differ from both inputs.

Gold-page recall is consistent with a coverage contribution to the loss: at $b = 5\%$, ColQwen2-fresh recalls 0.664/0.536 (per cell) vs. PAQ 0.494/0.443. This observation motivated the PAQ-T selection diagnostic and read-eval (§7.4); by itself it is not a causal decomposition of the accuracy gap.

Why the loss redirects rather than ends the project. Simply replacing mean aggregation with a better page-level attention head is not a clean novelty route: CAPS has already published expert-head attention selection on these benchmarks (§8.1). The PAQ-T read-eval (§7.4) has since shown that the expert head recovers the whole ≈ 10 -point gap — to *parity*, not past it — which is

EXPLORATORY: paq – gold_frequency_static by commitment pressure (formal interaction test includes 0 — not a confirmed mechanism)

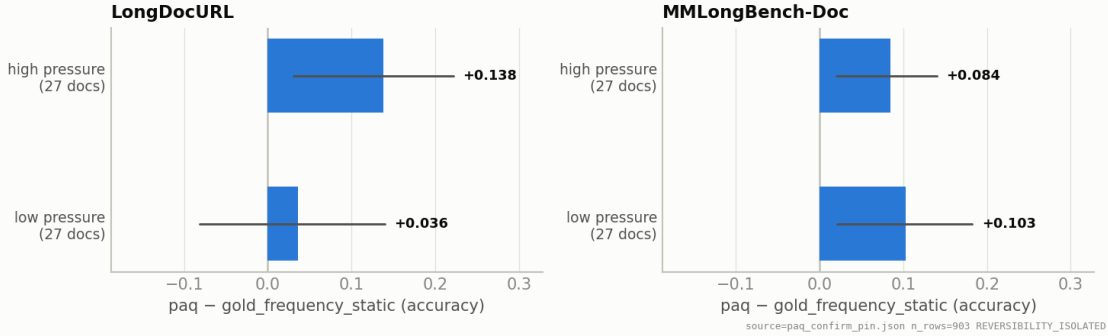


Figure 4: EXPLORATORY (demoted): PAQ–gold_frequency_static by commitment-pressure split. The formal high–low interaction includes zero on both contrasts; shown for completeness, not as a confirmed mechanism.

what redirected the program to the two lanes of §9.3.

7.3 Mechanism: demoted to exploratory

RETRACTED AS A CLAIM The pre-registered “commitment-pressure gradient” mechanism FAILED its formal test; what replaced it is a corrected, two-part account of where the benefit comes from.

We had hypothesized the reversibility benefit concentrates in high-commitment-pressure documents (Eq. (3)) and reverses under low pressure. The screen suggested this; the formal document-clustered interaction test on the confirmatory does **not** support it: high–low pressure difference +0.042 [−0.048, +0.130] on PAQ–static and −0.015 [−0.116, +0.080] on PAQ–paq_stale — both include zero. LongDocURL shows the predicted direction without significance; MMLongBench-Doc’s benefit is broad across pressure. Figure 4 is therefore labelled exploratory and this document makes no mechanism-gradient claim.

An earlier “reading quality, not coverage” explanation was also **wrong**: PAQ’s gold-recall exceeds the static’s by +0.092 (the earlier “recall $\approx$ static” countercheck was in error), and at *matched* recall ($n = 415$) PAQ still beats the static by +0.049 [+0.021, +0.076]. The corrected descriptive statement is: **per-query re-selection has higher gold coverage overall and higher downstream accuracy among queries with equal measured gold recall**. The matched-recall analysis does not identify why the selected page sets differ or establish a separate causal “reading quality” mechanism.

7.4 PAQ-T: accuracy read-eval — non-inferiority (parity), not a beat

EXECUTED — PARITY The pre-registered accuracy read-eval verdict is PAQ_T_ACC_NONINFERIOR: PAQ-T *matches* ColQwen2-fresh at matched page budget, training-free, with no separate model — but does **not** beat it. The head selection is benchmark-supervised (CV), and the expert-head technique is CAPS’s.

Recall gate (clean, crop-fixed). The PAQ-T gate (score-only; no reads) asked whether the recall deficit behind the −10pt loss is an artifact of mean aggregation and scoring resolution. Regen-

Table 5: PAQ-T page-selection recall@ $b=5\%$ diagnostic (*clean*, crop-fixed gate). Sample sizes differ slightly across rows, so they should not be read as paired deltas.

Scorer	LongDocURL	MMLongBench-Doc
coarse mean aggregation (PAQ v1)	0.494 ($n = 463$)	0.443 ($n = 440$)
higher-resolution mean aggregation	0.500 ($n = 451$)	0.452 ($n = 425$)
expert head (held-out-document CV)	0.683 ($n = 451$)	0.626 ($n = 425$)
ColQwen2 in-harness reference	0.664 ($n = 463$)	0.536 ($n = 440$)

Table 6: PAQ-T accuracy read-eval at $b=5\%$, near-full coverage (903 paired queries; document-cluster paired bootstrap; contrast rows Δ [LB₉₅, UB₉₅] where available). Verdict: PAQ_T_ACC_NONINFERIOR — both cells clear the pre-registered LB₉₅ > −0.03 margin; the BEATS bar (all LB₉₅ > 0) is *not* met.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
PAQ-T accuracy	0.518	0.357	—
ColQwen2-fresh accuracy	0.512	0.328	—
PAQ-T − ColQwen2-fresh	+0.006 [−0.018, +0.030]	+0.028 [+0.001, +0.055]	+0.017 [−0.000, +0.035]

erated clean after the KV fix (§4.3), its verdict is PAQ_T_VIABLE: held-out-document expert-head recall@ $b=5\%$ is **0.683** (LongDocURL, $n = 451$) and **0.626** (MMLongBench-Doc, $n = 425$) vs. ColQwen2’s in-harness 0.664 and 0.536 — PAQ-T recall meets or exceeds ColQwen2 on both cells (+1.9/+9.0pt). The expert head is the lever (mean aggregation: coarse 0.494/0.443, higher-resolution 0.500/0.452); resolution contributes almost nothing. Selected fold heads are 21_11/21_18 (LongDocURL) and 21_11/21_11 (MMLongBench-Doc), all layer 21 (Table 5).

The accuracy read-eval (pre-registered, frozen bars; protocol in §4.4). Table 6 gives the result at near-full coverage (463/463 LongDocURL with 12 OOM’d queries recovered; 440/444 MMLongBench-Doc with 19 recovered; the 4 *mi_phone* queries remain unpairable):

The missingness story matters and we tell it in full. The *complete-case* verdict (872 paired queries) looked slightly better: pooled +0.023 [LB₉₅ = +0.005], mmlongdoc +0.039 [+0.014]. The result cross-review (Sol) blocked it for **informative missingness**: the $\approx 4\%$ of queries whose high-res scoring OOM’d (6 big documents) have cached ColQwen2 accuracy 0.62 — far above the complete-case cell means (0.51/0.32) — so dropping them optimistically biased the comparison toward PAQ-T, and the adversarial imputation failed the non-inferiority bar. Recovering 31 of the 35 missing rows (conservative lower-res pass, §4.4) resolved the block and **erased the pooled edge**: the honest, near-full-coverage headline is the parity row above, with pooled LB₉₅ = −0.000. The reviewer was right; the complete-case numbers are retained only as this paragraph.

Secondary contrasts (near-full coverage, 872-query paired set). PAQ-T beats the original coarse PAQ by +0.119 [+0.090, +0.150] and high-resolution mean aggregation by +0.101 [+0.073, +0.132] — the expert head, not resolution, is the accuracy lever. PAQ-T sits only −0.035 [−0.065, −0.007] below full-context. ColQwen2-fresh > coarse-PAQ is reconfirmed at +0.097 [+0.069, +0.127].

What this is and is not (blunt). A training-free, single-model, reader-internal selector now *matches* a SOTA trained visual retriever on end-to-end accuracy at matched page budget, in the

amortized multi-query regime — with no separate retriever model and no multi-vector index. That is a real property, and it discharges the “within 2–3 points” non-inferiority bar predeclared by the efficiency red-team. It is **not** the beat this program is mandated to deliver: the pooled LB_{95} sits exactly at zero, LongDocURL is parity, and the mmlongdoc per-cell $LB_{95} = +0.001$ is a razor’s edge we refuse to sell as a win. Two caveats carry over: the head selection is benchmark-supervised CV (scope in §4.4), and the selector technique is CAPS’s — our residual novelty remains the amortized one-prefill-many-queries architecture.

7.5 Efficiency framing (measurement pending)

UNMEASURED No latency/VRAM/throughput table exists yet. Everything in this subsection is an accounting argument, flagged as such, with the red-team’s caveats folded in.

PAQ is training-free, uses no separate retriever model and no multi-vector index; the coarse pre-fill is paid once per document and each query costs one short forward over the retained cache plus the reread. But the honest accounting (from the efficiency red-team) is: ColQwen2 *also* amortizes (page embeddings are computed once; a query costs a small query encoding plus MaxSim), and PAQ’s per-query readout is a reader-scale forward — likely *more* per-query FLOPs than ColQwen2’s query encoding. What PAQ genuinely saves is *dependencies*: no second model resident, no index storage, one deployed artifact. Where the amortization argument has real teeth is against CAPS, which re-forwards every page per query (11.25s/query reported) and explicitly concedes the multi-query regime to embedding methods. An efficiency-only justification while losing ≈ 10 accuracy points was reviewed as weak-reject; we therefore make no efficiency claim until (a) the measured table exists (cold/warm start, retriever co-resident vs. sequential, index bytes/page, several queries-per-document), and (b) PAQ-T accuracy lands within the predeclared non-inferiority bars (within 2–3 points, $LB_{95} > -0.03$, both cells). Condition (b) is now met (§7.4: PAQ_T_ACC_NONINFERIOR); condition (a) — the measured table — still does not exist, so this subsection remains an accounting argument and no efficiency claim is made.

8 Related work

8.1 CAPS: the page-level method is already published

CAPS (Zhu et al., 2026) (“Attention as Selector”, ACL 2026) publishes attention-based page selection for long-document retrieval *on our exact benchmarks*: a small scoring VLM’s final-token→page attention at a single expert (layer, head), lightly calibrated on 6.4k off-benchmark contrastive triplets, beats ColPali-pipeline retrieval (MMLongBench R@5 78.89 vs. 72.00; downstream QA 44.69 vs. 36.18–41.01). Their training-free ablation lands just *below* ColPali (best zero-shot head R@5 70.92 vs. 72.00) — independently corroborating our own training-free loss. We state plainly: **the “reader-attention selects pages” method is scooped**, and any contribution of ours must be argued elsewhere. CAPS’s own concessions define the open ground: it scores every page independently per query (multi-query latency conceded to embedding methods — exactly PAQ’s amortized regime) and operates at page granularity only (region-level selection named as an open problem — exactly extension E1).

8.2 KV compression and eviction

H2O (Zhang et al., 2023), SnapKV (Li et al., 2024), FastV (Chen et al., 2024), LOOK-M (Wan et al., 2024), and KVzip (Kim et al., 2025) commit a compressed cache once; Quest (Tang et al.,

2024) shows query-aware beats query-agnostic at tight budgets in single-query text — motivation for, but not evidence of, our multi-query claims. SparseVILA (Anonymous, 2025d) is the nearest VLM neighbor for extension E2 (prefill pruning + decode-time retrieval) but prunes irreversibly at prefill and never compares against retrieval pipelines on long-document VQA; Kamera (Anonymous, 2026c) shows the multi-query KV-reuse serving regime is becoming real.

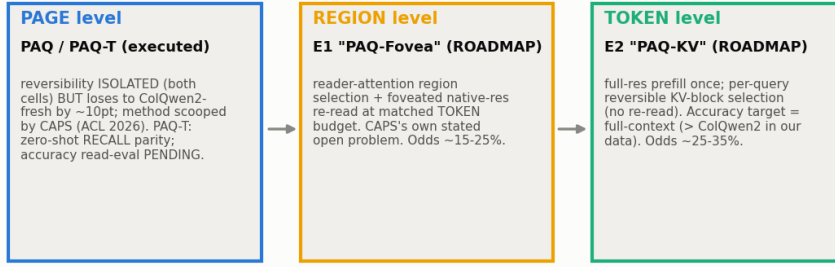
8.3 Trained visual retrieval

ColPali/ColQwen2 (Faysse et al., 2025) is the strong deployable retriever we lose to; M3DocRAG (Cho et al., 2024) and successors are the pipelines CAPS beat. The bar is moving: ColQwen3-class models and small distilled retrievers keep improving, so even a future win over ColQwen2 must be argued as a mechanism/regime result, not a leaderboard entry. A published oracle-analysis on MMLongBench (Anonymous, 2026b) mirrors our fusion finding (best-per-query routing between retrievers gains +0.9; union reranking gains more), and attention-based reranking precedents exist in text (Chen, Gutierrez, and Su, 2025; Wu et al., 2025).

8.4 Benchmarks and the region-level neighborhood

MMLongBench-Doc (Ma et al., 2024) and LongDocURL (Deng et al., 2025) supply the multi-query long-document structure; MMDocIR (Dong et al., 2025) adds layout-level retrieval labels useful for E1’s localization sanity checks. The region-level neighborhood is active — RegionRAG (Yuan et al., 2025) (trained retriever-side crops; near-flat on long docs, no long-doc QA), Snappy (Anonymous, 2025c) (training-free patch→OCR-box propagation, localization only), AGREE (Anonymous, 2025a) (reader attention distilled into a trained retriever), VRAG-RL (Wang et al., 2025) and DocV* (Anonymous, 2026a) (agentic crop/zoom loops), ViCrop/FOCUS (Zhang et al., 2025; Anonymous, 2025b) (single-image training-free cropping). In our current search we did **not identify prior work in the exact combined cell**: reader-internal attention as the runtime region selector, end-to-end QA at matched token budget vs. page-level retrieval, on long documents, multi-query amortized. This is a provisional literature-map claim, not proof of novelty; it motivates the E1 proposal. Page-selection-then-read itself long predates us (Tito, Karatzas, and Valveny, 2024).

The reversibility thesis pushed below page granularity — page **QUANTIZATION**, not selection quality, is the binding constraint



Region/token levels are PROPOSED extensions (~1-in-3 odds each), NOT results.

source=paq_confirm_pin.json n_rows=903 REVERSIBILITY_ISOLATED

Figure 5: The page → region → token reversibility roadmap. Page level is executed (v1 lost; PAQ-T parity; the method is CAPS's); the token level (PAQ-KV, Lane C) is in its development screen; the region level remains a proposal. Both active lanes are ~15–25% bets (§9.3). The unifying observation: retrieval quantizes evidence to whole pages, so at a fixed budget the open opportunity is below page granularity, where a reader that has already prefilled the document can re-select reversibly per query.

9 Current state, open questions, roadmap

9.1 Scoreboard (2026-07-16)

Item	Status	Where
Multi-query reversibility isolation (page level)	CONFIRMED (903 q, both cells)	§7.1
robustness to the KV-cache fix	VERIFIED (claim stands)	§4.3
Executed retrieval-vs-selection panel at 64K	DONE (first of its kind, to our knowledge)	§3, App. A
PAQ v1 vs. ColQwen2-fresh	LOST (−0.099 pooled)	§7.2
Page-level attention selection as a method	SCOOPED (CAPS, ACL 2026)	§8.1
Commitment-pressure mechanism	DEMOTED (interaction test incl. 0)	§7.3
KV-cache pollution bug	FOUND, QUANTIFIED, FIXED	§4.3
PAQ-T recall parity (clean gate)	CONFIRMED (0.683/0.626 vs. 0.664/0.536)	§7.4
PAQ-T accuracy read-eval	EXECUTED = PARITY (non-inferior, <i>not</i> a beat)	§7.4
Efficiency table	PENDING (unmeasured)	§7.5
Lane B: PAQ-T-LoRA off-benchmark adapter	DESIGNED + CPU prep done , GPU-gated, ~15–25%	§4.6, §9.3
Lane C: PAQ-KV below-page reversible KV	SPIKE PASSED; dev screen RUNNING , ~15–25%	§4.5, §9.3
Sealed fresh test set (73/45 docs)	RESERVED, UNTOUCHED (one-touch final eval)	§9.3
E1 PAQ-Fovea (region-level foveation)	ROADMAP (parked behind B/C)	§9.3

9.2 The thesis going forward

The reframe that survived every review round: **page quantization, not selection quality, is the binding constraint**. ColQwen2 wins the page-ranking game, CAPS owns the attention variant of it, and PAQ-T shows that fixing the selector buys back the whole gap — but only to parity. A whole-page budget is wasteful (most of a kept page is not evidence), and a page-level index cannot re-allocate budget within pages. The reader's retained prefill can (Figure 5).

9.3 The two-lane program to convert parity into a beat

The program mandate is explicit: ship a *method that beats the baselines*; a parity or boundary paper is not an acceptable terminus. After the read-eval, the human directed a two-lane program; both lanes were designed, cross-family reviewed (twice), and are being executed cheapest-risk-first.

Lane B — PAQ-T-LoRA off-benchmark selection-adapter (§4.6). Status: designed; two review rounds returned REVISE/HOLD then HOLD, with all blockers folded into the design (off-benchmark provenance, fully external selection, packed-context training, one deterministic resolution policy, adversarial imputation). CPU preparation is **done**: a **sealed fresh final test set** of 73 LongDocURL documents (433 queries) and 45 MMLongBench-Doc documents (244 queries), zero overlap with the 80-document pin, plus a provenance index (perceptual hash + source identity over all 40,442 benchmark pages) and the per-head mixture harness. All GPU stages (feature dump, adapter training, gates) are human-gated. Known weakness, disclosed: the fresh mmlongdoc cell (244 queries/45 documents) sits below the program’s 250-query power floor, so a beat there needs a larger effect or more documents.

Lane C — PAQ-KV below-page reversible KV selection (§4.5). Status: designed with CPU pricing from the caches; the decode-identity spike **passed** (the mask path is exact: zero mask-specific argmax changes); a 30-document/cell development screen with frozen STOP gates is **running at the time of writing**. We report no screen numbers here — by design, an in-progress run is not a result. The lane’s computed headwind is stated up front: the cached confirmatory prices full-context – ColQwen2-fresh at $LB_{95} = -0.001$ on LongDocURL (vs. $+0.032$ on MMLongBench-Doc), so on LongDocURL even a *perfect* selector that exactly matches full-context fails the beat bar — the win there requires *denoising above* full-context (the room exists: oracle–full $+0.064$, $LB_{95} = +0.032$, on that cell; whether block-level selection captures it is exactly what the screen tests).

The sealed-test discipline (program-level, adopted from review). The 80-document confirmatory pin has been reused across this program and is therefore **development-only** from now on, for both lanes. The fresh 73/45-document set is **sealed for a single one-touch final evaluation**: ColQwen2 will be run on it once, and the only mandate-grade claim this program will make is a both-cell, adversarially-imputed, document-clustered $LB_{95} > 0$ on that sealed set. *The sealed set has not been evaluated; no number in this document comes from it.*

The honest shared risk. PAQ-T already has *higher* gold-page recall *and* precision than ColQwen2 on identical budgets, yet lands at accuracy parity. Stratified by who wins the per-query selection contest, selection quality does convert to accuracy ($\approx \pm 0.2$ /query: PAQ-T wins selection on 141 queries $\rightarrow +0.206$; loses on 61 $\rightarrow -0.249$; ties on 670 $\rightarrow +0.008$) — parity is a *cancellation*, not a conversion failure. But the aggregate pattern still admits the pessimistic reading that the reader’s recall $\rightarrow$ accuracy **conversion is the true wall**, in which case both lanes are polishing selection while the binding constraint is elsewhere. For this reason we hold both lanes at **$\sim 15\text{--}25\%$** for the mandated beat, and say so. E1/PAQ-Fovea (region-level foveation, CAPS’s own named open problem) is parked behind B/C. If neither lane survives its gates, the honest terminus of this program is what is already in this document: the reversibility regime result, the executed panel, the parity result, and the boundary map.

9.4 Open questions

(1) Does the C-lane development screen clear its frozen gates — in particular ceiling transfer and the LongDocURL denoise-above-full guard? (2) Does an off-benchmark-trained adapter transfer (Lane B’s cheap gate), and can it grow PAQ-T’s selection wins rather than merely fixing losses (moving all 61 loss-queries to ties is worth only $\approx +1.8\text{pp}$ pooled)? (3) Is the reader’s recall $\rightarrow$ accuracy conversion the true wall (the shared-risk reading above)? (4) Does token-level selection preserve full-context accuracy on multi-hop/cross-page questions (Quest-family methods degrade exactly there, and MMLongBench-Doc is 33% cross-page)? (5) The moving bar: ColQwen3-class retrievers may outdate any ColQwen2 comparison by submission time — any win must be a regime/mechanism claim. (6) A second VLM (currently everything is a Qwen3-VL-8B case study).

10 Limitations

- **Heterogeneity.** Effect sizes differ by cell (e.g. the stale gap is $+0.257$ vs. $+0.163$); we never average cells except in the labelled pooled statistic, and per-cell results are primary.
- **Page routing only.** v1 selects pages and rereads at full resolution; it is not KV compression, makes no memory/latency claim, and the systems-level story is unmeasured (§7.5).
- **Base-model ceiling.** 35.1% of confirmatory queries are oracle-unanswerable (score 0 given exactly the gold pages); only 45.1% are fully oracle-answerable. Absolute numbers are deflated for every arm; oracle is not a strict ceiling (28 PAQ wins over it).
- **Resolution cap.** The high-res vision encoder OOMs on many-page documents, so PAQ-T scoring was capped at a 6,000-token target (and the read-eval’s recovery pass used a reduced 4,000-token target on 6 large documents, conservative against PAQ-T) — the resolution lever is under-tested.
- **CAPS overlap.** Expert-head attention selection is CAPS’s published method; any PAQ-T claim is limited to the amortized multi-query architecture and must credit CAPS.
- **PAQ-T parity caveats.** The head selection is cross-validated, *benchmark-supervised* model selection (not zero-shot), and the bootstrap does not propagate the CV head-selection uncertainty; the mmlongdoc per-cell $\text{LB}_{95} = +0.001$ is a razor’s edge and we do not present it as a per-cell win. Parity, not a beat, is the claim.
- **KV-cache bug residue.** The confirmatory run predates the crop fix (§4.3); the isolation *contrasts* were verified robust, but the absolute PAQ accuracy levels in the confirmatory carry ≈ 0.04 inflation on the 24-document check and have not been re-run clean at full scale.
- **Weak family ports.** FastV/LOOK-M/KVzip are disclosed proxies and SnapKV is a weak port; we claim nothing against canonical versions of them.
- **Single reader.** All results are a Qwen3-VL-8B case study; generalization to other VLMs is untested.
- **Stale-policy dependence.** The commit-once aggregate uses the first $M = \min(3, k)$ queries in frozen arrival order; order-permutation and M -sensitivity robustness checks were reviewer-requested and remain to be run (the gold-informed static, which is order-free, partially covers this).

A Full executed panel at the gate budget

Table 7: Every arm, accuracy at $b=5\%$, per cell (verbatim from `research/paq_verdict.json`). The query-blind family sits at/below random gold-recall — the anti-strawman record: beating it is visibly not the claim.

Group	Arm	LongDocURL	MMLongBench-Doc
ours	PAQ	0.386	0.258
	attn_max (alt-agg)	0.337	0.152
isolation	paq_stale (commit-once)	0.128	0.095
	gold_frequency_static	0.278	0.154
deployable retrieval	ColQwen2-fresh	0.512	0.328
	ColQwen2-stale	0.216	0.138
	CLIP-fresh	0.155	0.164
reference (not competitors)	SnapKV-fresh	0.298	0.157
	full-context	0.546	0.397
	oracle-evidence (not strict)	0.609	0.431
query-blind controls	SnapKV-stale	0.083	0.074
	H2O	0.045	0.039
	FastV (ctx-only proxy)	0.045	0.031
	LOOK-M ($\rightarrow$ H2O on images)	0.045	0.039
	KVzip (recv_max proxy)	0.047	0.048
	random	0.068	0.022
	truncation	0.033	0.041

Secondary statistics (pooled, reported not gated): PAQ beats the best family arm (SnapKV-stale) with min-over-family $LB_{95} = +0.202$; $PAQ - SnapKV\text{-}fresh = +0.094$ [$+0.069, +0.122$]; $PAQ - attn_max = +0.077$ [$+0.048, +0.110$]; $PAQ - oracle = -0.199$ [$-0.239, -0.161$].

Gold-page recall at $b=5\%$ (per cell, LongDocURL / MMLongBench-Doc): PAQ 0.494/0.443; gold_frequency_static 0.381/0.373; paq_stale 0.137/0.148; ColQwen2-fresh 0.664/0.536.

B The adversarial review trail (anti-strawman record)

This project’s positives were only believed after surviving independent cross-family review; two of them did not survive, and we list the retractions because they are part of the evidence:

1. **Pre-freeze design review** (code-faithfulness + methodology, 2026-07-14): folded best-of-family gating, document-cluster bootstrap, the page-routing estimand disclosure, first- M stale aggregation, and power floors *before* any GPU run.
2. **v1 gate retraction.** The first gate (PAQ_V1_G0, PAQ beats the eviction family) *passed and was then retracted*: reviewers + an independent reproduction showed a gold-informed static nearly matched PAQ’s recall — the family’s weakness was a weak-static artifact, and reversibility had not been isolated. The two-comparator isolation gate (Eq. (6)) replaced it and was frozen before the re-run.
3. **Result review of the isolation positive** (2026-07-14): heterogeneity honestly surfaced (screen-scale MMLongBench static contrast -0.017 , CI incl. 0 — later resolved at 80 docs); `attn_best`, a derived arm that peeked at gold task score, was dropped.

4. **Final AAAI-readiness review** (2026-07-15): named the missing deployable retrieval baseline as the #1 threat — running it produced the decisive loss (§7.2); forced the `best_static` → `gold_frequency_static` rename and the `LB95` relabel; demoted the mechanism claim (§7.3); required the independent-extension and missingness analyses (§7.1).
5. **Efficiency red-team** (2026-07-15): rejected efficiency-only framing (§7.5) and predeclared the non-inferiority bars PAQ-T must meet.
6. **Read-eval BUILD review** (2026-07-16, cross-family): found the **KV-cache pollution bug** (§4.3) as a blocker — the fix, the blast-radius diagnostic, the clean regeneration of the recall gate, and the reversibility robustness check all followed from it; also forced the gate-pinned auditable fold map, per-cell power floors, per-cell missingness sensitivity, and the reframing of the claim as “cross-validated, benchmark-supervised (not zero-shot).”
7. **Read-eval RESULT review** (2026-07-16): one reviewer passed the numbers (hashes, bars, pairing verified); the other **blocked** the complete-case verdict for *informative missingness* (the OOM’d queries were easy for the baseline). The block was resolved by recovering 31/35 missing rows conservatively; the recovery erased the pooled edge (§7.4). This is the program’s clearest example of a result review changing a headline.
8. **Lane-B design reviews** (2026-07-16, two rounds: REVISE/HOLD then HOLD): dropped a benchmark-contaminated training source (MoLoRAG), forced fully external selection of every design choice, a fresh held-out document set, 100% coverage with adversarial imputation, packed-context training, an adapter-off invariance test, and honest odds of ~15–25%.
9. **Combined B+C review and C screen decision** (2026-07-16): imposed the program-level **sealed-test discipline** (the 80-document pin is development-only; the fresh 73/45 set is sealed for one touch); corrected the C-lane spike design (M-RoPE position continuation; the “nominal-5%” budget relabel); and gated the C screen on a decode-identity audit as a kill criterion — which then passed with zero mask-specific argmax changes (§4.5).

C Reproducibility

- **Harness** (frozen): `scripts/paq.py` (v1 sha256 735eeca..., commit ca66292; v2 e9e06aad..., commit f7e7a53; v3 = v2 + confirmatory pin, re-scored at commit b653e85). Retrieval arms: `scripts/paq_retrieval_arms.py` (ColQwen2 vidore/colqwen2-v1.0 + CLIP; incremental merge — the frozen gate math and the retrieval-free rows are untouched). PAQ-T gate: `scripts/paq_t_gate.py`. Figures: `scripts/paq_figs.py` (all numbers read from the verdict JSONs; every figure carries a source stamp `pin.n_rows.verdict`).
- **Pins**: screen `research/paq_pinned_subset.json` (sha256 b194ed84..., chmod 444, byte-verified by the harness); confirmatory `research/paq_confirm_pin.json` (80 docs/cell; the screen pin is its prefix).
- **Verdicts**: `research/paq_verdict.json` (gate + budget curves + retrieval secondary + strata), `research/paq_t_gate_verdict.json` (clean, crop-fixed); per-query per-arm rows in `research/paq_results.jsonl`; cached retrieval scores in `research/paq_retrieval_scores.json`.
- **Read-eval + integrity artifacts (2026-07-16)**: harness `scripts/paq_t_readeval.py` with pre-registered bars in `research/paq_t_readeval_frozen_meta.json` (frozen, sha-pinned, before the run); verdicts `research/paq_t_readeval_verdict.json` (complete-case) and `research/paq_t_readeval_combined_verdict.json` (near-full coverage; the headline); recovery `scripts/paq_t_readeval_recover.py`. KV-bug artifacts: blast radius `research/paq_kvcrop_diag.json`, reversibility robustness `research/paq_revcrop_verdict.json` (`scripts/paq_reversibility_crop_check.py`).

- **Lane artifacts (2026-07-16):** C-lane design `research/design_notes/paqc_e2_paqkv_design_2026-07-16.md`, spike `scripts/paqc_spike.py` with `research/paqc_spike_test{A,B}.json` and `research/paqc_identity_audit.j`. B-lane design `research/design_notes/paqt_lora_calibration_design_2026-07-16.md` (incl. the folded review addenda) and data/harness spec `research/design_notes/paqb_data_harness_spec_2026-07-16.md` (`scripts/paqb_prep.py`; sealed fresh pin + provenance index). Review archive: `research/reviews/paqt_2026-07-16`
- **Gate + bootstrap + pairing** are unit-tested (`paq.py --selftest`); the belief gate, arm classifications, and strata were pre-registered in the frozen bar.
- **Commands:** `gate python scripts/paq.py --run --cell vision --gate-only`; full contest `--run --cell all`; scoring `--score`; figures `python scripts/paq_figs.py`.

D Timeout recovery note

The confirmatory hit its 10-hour wall-clock limit during final scoring (exit 124) at 903/907 vision queries. Because the harness flushes each row as it completes, the run was recovered by re-scoring the flushed rows on CPU (no GPU re-run, no selection re-computation); the 4 unscored queries are the `mi_phone` rows of §7.1, and the worst-case imputation leaves every gate conclusion unchanged. A separate single-query text cell (“counting”) was truncated by the same timeout and is out of scope for the multi-query claims; it was not re-run.

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